AI-Powered Due Diligence & Compliance Risk Intelligence Platform

Product Requirements Document (PRD)

MSBA Portfolio Project | Data Mining, AI, Analytics and RAG

Document Type	Product Requirements Document
Project Theme	Due Diligence, Compliance Risk, AML/KYC Analytics
Primary Audience	MSBA portfolio reviewers, compliance stakeholders, analysts and product evaluators
Author	Sudhanshu Mathur
Version	1.0
Date	25 June 2026

Educational and Portfolio Use Notice

This document describes a synthetic-data analytics project for educational and portfolio purposes. It is not intended for real regulatory decisions, legal advice, or production compliance operations. Real bank, client, or confidential information should not be used.

Prepared as a business analytics and AI-assisted decision-support project, not as a production compliance system.

Table of Contents

1. Product Overview 7

1.1 Product Name . . . 7

1.2 One-Line Description . . . 7

1.3 Product Vision . . . 7

2. Background and Business Context 7

2.1 Existing Manual Process . . . 7

2.2 Business Problem . . . 7

2.3 Business Opportunity . . . 8

3. Goals and Objectives 8

3.1 Primary Product Goals . . . 8

3.2 MSBA Project Goals . . . 8

3.3 Non-Goals . . . 8

4. Target Users and Personas 9

4.1 Primary Users . . . 9

4.2 Secondary Users . . . 9

4.3 Personas . . . 9

5. Product Scope 9

5.1 MVP Scope . . . 9

5.2 Version 2 Scope . . . 9

5.3 Out of Scope for MVP . . . 10

6. End-to-End Product Workflow 10

6.1 High-Level Workflow . . . 10

6.2 Case Lifecycle Workflow . . . 10

6.3 Final Case Outcomes . . . 10

7. Data Requirements 11

7.1 Data Strategy . . . 11

7.2 Recommended Dataset Size . . . 11
7.3 Client Master Data . . . 11
7.4 Ownership and Board Data . . . 12
7.5 Screening Hits Data . . . 12
7.6 Country Risk Data . . . 12
7.7 Adverse Media Data . . . 13
7.8 Previous Case History . . . 13
8. Data Mining Requirements 13
8.1 Data Ingestion . . . 13
8.2 Data Cleaning . . . 13
8.3 Data Quality Rules . . . 14
8.4 Data Mining Output . . . 14
9. Entity Matching Requirements 14
9.1 Purpose . . . 14
9.2 Matching Inputs . . . 14
9.3 Matching Logic . . . 15
9.4 Match Score Bands . . . 15
9.5 Functional Requirements . . . 15
9.6 Entity Matching Output . . . 15
10. Risk Scoring Requirements 15
10.1 Purpose . . . 15
10.2 Risk Factors and Suggested Weights . . . 16
10.3 Risk Level Bands . . . 16
10.4 Recommended Actions . . . 16
10.5 Functional Requirements . . . 16
10.6 Example Risk Explanation . . . 17
11. AI and NLP Requirements 17
11.1 Purpose . . . 17
11.2 AI Capabilities . . . 17

11.3 Named Entity Recognition . . . 17
11.4 Adverse Media Classification . . . 17
11.5 AI Output Example . . . 18
11.6 Functional Requirements . . . 18
11.7 AI Guardrails . . . 18
12. RAG Requirements 18
12.1 Purpose . . . 18
12.2 Knowledge Base . . . 18
12.3 RAG Workflow . . . 19
12.4 Example Analyst Questions . . . 19
12.5 RAG Answer Requirements . . . 19
12.6 Example RAG Answer . . . 19
12.7 Functional Requirements . . . 20
13. Analytics Dashboard Requirements 20
13.1 Purpose . . . 20
13.2 Dashboard Page 1: Executive Overview . . . 20
13.3 Dashboard Page 2: Case Queue . . . 20
13.4 Dashboard Page 3: Client 360 View . . . 20
13.5 Dashboard Page 4: Country Risk Exposure . . . 21
13.6 Dashboard Page 5: Analyst Productivity . . . 21
13.7 Functional Requirements . . . 21
14. Due Diligence Report Requirements 21
14.1 Purpose . . . 21
14.2 Report Sections . . . 21
14.3 Example Report Summary . . . 22
14.4 Functional Requirements . . . 22
15. Case Management Requirements 23
15.1 Case Statuses . . . 23

15.2 Case Actions . . . 23
15.3 Functional Requirements . . . 23
16. Human-in-the-Loop Requirements 23
16.1 Purpose . . . 23
16.2 Human Review Points . . . 23
16.3 Analyst Override . . . 24
16.4 Functional Requirements . . . 24
17. Model and Analytics Evaluation 24
17.1 Purpose . . . 24
17.2 Evaluation Metrics . . . 24
17.3 Baseline Comparison . . . 24
17.4 Success Criteria . . . 24
18. Non-Functional Requirements 25
19. Technical Approach 25
19.1 Recommended Stack for Non-Coder / MSBA Project . . . 25
19.2 Simplified Architecture . . . 25
19.3 Optional Advanced Architecture . . . 25
20. Key Screens and Pages 26
21. User Stories and Acceptance Criteria 26
22. MVP Deliverables 26
23. Project Roadmap 27
24. Success Metrics 27
24.1 Product Success Metrics . . . 27
24.2 Business Impact Metrics . . . 27
24.3 Academic / Portfolio Metrics . . . 27
25. Risks and Mitigations 28
26. Assumptions 28

27. Open Questions 28

28. Recommended MVP for MSBA 28

29. Final Product Positioning 29

30. Appendix: Glossary 29

1. Product Overview

1.1 Product Name

AI-Powered Due Diligence & Compliance Risk Intelligence Platform.

1.2 One-Line Description

A data-driven compliance analytics platform that consolidates client, ownership, screening, country-risk, and adverse media data to support due diligence, explainable risk scoring, case prioritization, RAG-based investigation, and automated report generation.

1.3 Product Vision

Compliance and due diligence teams often spend significant time manually reviewing client profiles, screening alerts, ownership structures, political exposure, adverse media, sanctions exposure, country risk, and internal case history across multiple Excel files and platforms.

This platform creates a unified risk intelligence system that helps analysts identify high-risk clients faster, understand the drivers behind risk classifications, prioritize cases, and generate evidence-backed due diligence reports.

Human-in-the-loop principle

The platform does not replace compliance analysts. It supports human decision-making by providing structured evidence, explainable risk scores, analytics dashboards, and AI-assisted summaries.

2. Background and Business Context

2.1 Existing Manual Process

In a typical KYC, AML, and due diligence workflow, analysts receive alerts or hits related to clients, companies, directors, shareholders, beneficial owners, or associated entities.

- Sanctions screening
- PEP screening and political association review
- Adverse media monitoring
- Internal watchlist hits
- Regulatory findings
- High-risk jurisdiction exposure
- Previous investigation history

Analysts then manually review client profiles, company details, board members, shareholding and beneficial ownership, financial information, country exposure, current relationship status, previous alerts, case notes, internal platform status, and public research sources.

2.2 Business Problem

- Data is spread across multiple files and systems.
- Analysts spend significant time searching, validating, and reconciling client data manually.
- False positives are common in name screening because similar names can generate irrelevant alerts.
- Risk scoring can be inconsistent across analysts or teams.
- Case prioritization is difficult when alert volumes are high.
- Reports are manually written and may vary in quality, depth, and structure.
- Evidence used for decisions may not be easily traceable.
- Management often lacks a consolidated analytics view of compliance risk.

2.3 Business Opportunity

- Consolidate client due diligence data into a single client risk profile.
- Identify risk indicators automatically from structured and unstructured data.
- Improve false-positive triage through entity matching and evidence checks.
- Prioritize cases based on explainable risk scores and urgency.
- Generate standardized due diligence summaries.
- Improve visibility into country, industry, and alert-type risk trends.
- Support analysts through a RAG-based investigation assistant that cites evidence.

3. Goals and Objectives

3.1 Primary Product Goals

- Consolidate multiple due diligence data sources into a unified client profile.
- Detect and classify compliance risk indicators.
- Perform entity matching to identify possible true hits and false positives.
- Calculate an explainable compliance risk score.
- Prioritize cases based on risk level and urgency.
- Provide analytics dashboards for business insights.
- Use RAG to support analyst investigation with grounded answers.
- Generate automated due diligence reports.
- Track case lifecycle from alert creation to final decision.
- Demonstrate end-to-end data mining, analytics, AI, and decision support.

3.2 MSBA Project Goals

- Business problem understanding and process mapping.
- Data mining and data preparation from multiple sources.
- Risk analytics and explainable decision modeling.
- Dashboard storytelling for stakeholders.
- AI-assisted decision support using NLP and RAG.
- Clear communication of insights to non-technical stakeholders.

3.3 Non-Goals

- The system will not replace compliance analysts.
- The system will not make final regulatory decisions automatically.
- The MVP will not use real bank or client confidential data.
- The MVP will not be a production-grade compliance system.
- The MVP will not provide legal, regulatory, or financial advice.
- The MVP will not connect to actual live sanctions databases.
- The MVP will not handle real-time transaction monitoring.

4. Target Users and Personas

4.1 Primary Users

User	Responsibilities	Primary Needs
Compliance Analyst	Reviews alerts, validates hits, checks evidence, determines risk, and prepares investigation reports.	Unified client profile, risk indicators, hit matching, evidence summary, report generation, case tracking.
Due Diligence Reviewer / Senior Analyst	Validates analyst findings and approves decisions.	Risk score explanation, case history, evidence trail, analyst notes, escalation summary.
Compliance Manager	Monitors workload, risk distribution, case aging, and team productivity.	Dashboard view, case queue, risk heatmaps, workload metrics, high-risk exposure analysis.

4.2 Secondary Users

User	Needs
Business Stakeholder	Understand whether a client can continue the relationship, requires monitoring, or needs restrictions.
MSBA Reviewer / Professor / Interviewer	Evaluate business problem clarity, analytics pipeline, AI/RAG demonstration, measurable impact, and storytelling quality.

4.3 Personas

Persona	Context	Goal	Success Criteria
Priya - Compliance Analyst	Reviews many alerts and switches between Excel files and internal trackers.	Quickly understand whether an alert is relevant and what evidence supports the decision.	Opens one client view, sees risk indicators, reviews evidence, and generates a first-draft report.
Arjun - Senior Reviewer	Reviews escalated cases and needs a clear reason for high-risk classification.	Validate evidence and approve, reject, or request more information.	Sees explainable scoring, evidence, analyst notes, and final recommendation.
Meera - Compliance Manager	Needs visibility into case aging, false positives, high-risk countries, and workload.	Monitor operational performance and risk trends.	Uses dashboards to identify bottlenecks and risk concentration.

5. Product Scope

5.1 MVP Scope

- Synthetic data generation or upload.
- Data ingestion from CSV or Excel files.
- Data cleaning and normalization.
- Client 360 profile creation.
- Entity matching using fuzzy matching.
- Rule-based risk scoring.
- Risk level classification.
- Case prioritization dashboard.
- RAG-based investigation assistant.
- Automated due diligence report generation.
- Case lifecycle tracking.
- Basic model evaluation metrics.

5.2 Version 2 Scope

- Advanced NLP classification for adverse media and analyst notes.

- ML-based risk prediction using historical outcomes.
- Network graph for ownership and association relationships.
- Scenario analysis and what-if simulations.
- Analyst feedback loop for improving scoring logic.
- Automated web research for public adverse media.
- Integration with public sanctions or PEP sample sources.
- Role-based access control, audit logs, OCR and document upload.

5.3 Out of Scope for MVP

- Real client data or confidential bank data.
- Live bank integrations or live regulatory screening.
- Legal or regulatory decision automation.
- Production-grade authentication or authorization.
- Transaction monitoring and real-time streaming alerts.
- Complex enterprise model governance workflows.

6. End-to-End Product Workflow

6.1 High-Level Workflow

```
Data Sources
    -> Data Ingestion
    -> Data Cleaning and Normalization
    -> Client 360 Profile Creation
    -> Entity Matching
    -> Risk Indicator Detection
    -> Explainable Risk Scoring
    -> Case Prioritization
    -> Analytics Dashboard
    -> RAG Investigation Assistant
    -> Due Diligence Report Generation
    -> Human Review and Final Decision
```

6.2 Case Lifecycle Workflow

```
New Alert
    -> Data Enrichment
    -> System Risk Assessment
    -> Analyst Review
    -> Classification
    -> Reviewer Approval
    -> Final Decision
    -> Action Tracking
    -> Case Closure
```

6.3 Final Case Outcomes

- False Positive
- True Hit
- Possible Match
- No Material Risk
- Medium Risk - Monitor
- High Risk - Enhanced Due Diligence

- Critical Risk - Escalate
- Restrict Relationship
- Quarantine Business Activity
- Offboard Client

7. Data Requirements

7.1 Data Strategy

The project should use synthetic data or publicly available sample data. Real bank data, real client data, and confidential information must not be used. The synthetic dataset should be realistic enough to reflect actual compliance workflows.

7.2 Recommended Dataset Size

Dataset	Recommended Volume
Clients	500 to 1,000 clients
Directors / Board Members / UBOs	1,500 to 3,000 records
Screening Hits	1,000 to 2,000 records
Adverse Media Articles	200 to 500 articles
Previous Case Records	100 to 250 records
Country Risk Records	100+ countries
Internal Watchlist Records	20 to 50 records

7.3 Client Master Data

Purpose: Stores the main client-level information used to create the Client 360 view.

Field	Description
client_id	Unique client identifier.
client_name	Legal name of the client.
client_type	Individual, Corporate, Trust, Fund, or other relevant type.
registration_number	Company registration or identity reference.
incorporation_country	Country of incorporation or residence.
operating_countries	Countries where the client operates or has exposure.
industry	Client industry or business activity.
onboarding_date	Date client was onboarded.
client_status	Active, Offboarded, Quarantined, Restricted.
relationship_status	Existing, New, Dormant.
annual_revenue	Approximate annual revenue.
risk_rating_existing	Existing risk rating where available.

7.4 Ownership and Board Data

Field	Description
entity_id	Unique person or entity identifier.
client_id	Linked client identifier.
entity_name	Name of director, shareholder, signatory, or beneficial owner.
role	Director, Shareholder, UBO, Signatory, or related party.
ownership_percentage	Ownership share where applicable.
nationality	Person nationality.
residence_country	Country of residence.
date_of_birth	DOB where applicable and available.
pep_flag	Whether entity is linked to political exposure.
sanction_flag	Whether entity has sanction exposure.
adverse_media_flag	Whether entity has adverse media exposure.

7.5 Screening Hits Data

Field	Description
hit_id	Unique hit identifier.
client_id	Linked client identifier.
matched_entity_name	Name from screening source.
internal_entity_name	Client/entity name in internal data.
hit_type	Sanction, PEP, Adverse Media, Internal Alert, Regulatory.
hit_source	Source of alert.
hit_date	Date alert was generated.
hit_country	Country associated with the hit.
severity	Low, Medium, High, Critical.
initial_status	New, Reviewed, Escalated.
source_reference	Document, link, or reference ID.

7.6 Country Risk Data

Field	Description
country	Country name.
country_risk_level	Low, Medium, High, Critical.
sanctions_exposure	Yes/No flag for sanctions exposure.
fatf_style_risk	Low, Medium, High risk bucket.
corruption_index_bucket	Low, Medium, High bucket.
conflict_zone_flag	Yes/No flag for conflict-zone association.
regulatory_risk_flag	Yes/No flag for regulatory risk.

7.7 Adverse Media Data

Field	Description
article_id	Unique article identifier.
client_id	Linked client if known.
entity_mentioned	Company or person mentioned in the article.
article_title	News headline.
article_text	Full or summarized article text.
source	Source name.
publication_date	Date published.
risk_category	Fraud, Bribery, Corruption, Money Laundering, Regulatory, No Risk.
sentiment	Negative, Neutral, Positive.
severity	Low, Medium, High, Critical.

7.8 Previous Case History

Field	Description
case_id	Unique case ID.
client_id	Linked client identifier.
previous_hit_type	Type of previous alert.
previous_decision	False Positive, True Hit, Escalated, Cleared.
analyst_notes	Previous analyst notes.
final_action	Continue, Monitor, Escalate, Offboard.
closure_date	Date case was closed.
reviewer_comments	Senior reviewer comments.

8. Data Mining Requirements

8.1 Data Ingestion

The system should allow the user to upload or load multiple CSV/Excel files and validate required columns before processing.

ID	Requirement	Priority
DM-001	System should accept CSV/Excel data files.	Must Have
DM-002	System should validate required columns.	Must Have
DM-003	System should show errors for missing required fields.	Must Have
DM-004	System should combine multiple files using client_id.	Must Have
DM-005	System should generate a unified client risk dataset.	Must Have

8.2 Data Cleaning

- Standardize country names.
- Remove leading and trailing spaces.
- Convert dates to a standard format.
- Normalize company suffixes such as Ltd, Limited, LLC, GmbH.
- Standardize hit type names.
- Remove duplicate records.

- Handle missing values.
- Validate ownership percentage.
- Flag inconsistent client status.

ID	Requirement	Priority
DM-006	System should normalize text fields.	Must Have
DM-007	System should standardize countries and dates.	Must Have
DM-008	System should detect duplicate clients and entities.	Must Have
DM-009	System should flag missing critical fields.	Must Have
DM-010	System should produce a data quality summary.	Must Have

8.3 Data Quality Rules

Issue	Example
Missing client name	client_name is blank.
Missing country	incorporation_country is blank.
Duplicate client	Same registration number appears twice.
Conflicting status	Client marked Active in one file and Offboarded in another.
Invalid ownership	Ownership percentage greater than 100.
Missing hit type	Screening hit exists without a type.
Missing source	Adverse media article has no source.
Invalid severity	Severity value outside allowed list.

8.4 Data Mining Output

The output of the pipeline should be a unified client profile containing client details, ownership details, board members, country exposure, screening hits, adverse media records, previous case history, risk indicators, risk score, and recommended action.

9. Entity Matching Requirements

9.1 Purpose

Entity matching helps determine whether a screening hit is likely related to the actual client, director, shareholder, or beneficial owner. This is important because AML/KYC screening often generates false positives due to similar names.

9.2 Matching Inputs

- Screening name
- Internal client/entity name
- Country
- Date of birth
- Registration number
- Role in company
- Nationality
- Ownership link

9.3 Matching Logic

```
Screening Name: Mohammed Ali Khan
Internal Name: Mohammad A. Khan
Similarity Score: 88%
Classification: Possible Match
```

9.4 Match Score Bands

Score Range	Classification
0-60	Likely False Positive
61-85	Possible Match
86-100	Potential True Match

9.5 Functional Requirements

ID	Requirement	Priority
EM-001	System should calculate similarity score between screening name and internal name.	Must Have
EM-002	System should classify match as false positive, possible match, or potential true match.	Must Have
EM-003	System should use additional fields such as country and DOB where available.	Should Have
EM-004	System should explain why a match was classified a certain way.	Must Have
EM-005	System should allow analyst override.	Should Have

9.6 Entity Matching Output

- Matched name
- Internal name
- Similarity score
- Country match status
- DOB match status
- Entity role
- Match classification
- Explanation

10. Risk Scoring Requirements

10.1 Purpose

The risk scoring engine converts risk indicators into an explainable compliance risk score. The score should not be a black-box output. Analysts should understand why a client received a particular score.

10.2 Risk Factors and Suggested Weights

Risk Factor	Score
Sanctions hit	+100
PEP association	+60
Adverse media - high severity	+50
Adverse media - medium severity	+30
High-risk country exposure	+30
Critical-risk country exposure	+50
Complex ownership structure	+25
Internal alert	+20
Previous true hit	+40
Previous false positive	-10
Active relationship	+10
Offboarded client	-20

10.3 Risk Level Bands

Score Range	Risk Level
0-40	Low
41-70	Medium
71-100	High
100+	Critical

10.4 Recommended Actions

Risk Level	Recommended Action
Low	Continue Relationship
Medium	Monitor
High	Enhanced Due Diligence
Critical	Escalate / Restrict / Offboard

10.5 Functional Requirements

ID	Requirement	Priority
RS-001	System should calculate risk score for each client.	Must Have
RS-002	System should classify clients into risk levels.	Must Have
RS-003	System should show score breakdown by risk factor.	Must Have
RS-004	System should recommend an action based on score.	Must Have
RS-005	System should allow scenario-based adjustment of risk factors.	Could Have
RS-006	System should allow analyst override with reason.	Should Have

10.6 Example Risk Explanation

```
Client: ABC Global Holdings
Total Risk Score: 92
Risk Level: High

Score Breakdown:
+50 Adverse media related to bribery
+30 High-risk country exposure
+20 Internal alert
-8 Previous false positive history adjustment

Recommendation:
Enhanced Due Diligence
```

11. AI and NLP Requirements

11.1 Purpose

AI/NLP should help analyze unstructured data such as adverse media articles, analyst notes, and previous case comments. The AI layer assists with classification, extraction, explanation, and summarization.

11.2 AI Capabilities

- Named entity extraction
- Adverse media classification
- Risk category detection
- Sentiment or severity classification
- Summary generation
- Hit explanation
- Report drafting

11.3 Named Entity Recognition

- Company names
- Person names
- Countries
- Political terms
- Crime-related terms
- Regulatory bodies
- Dates
- Monetary values

11.4 Adverse Media Classification

- Fraud
- Bribery
- Corruption
- Money laundering
- Terror financing
- Sanctions evasion
- Regulatory violation

- Tax evasion
- No material risk

11.5 AI Output Example

```
Input:
ABC Holdings is being investigated for alleged bribery in a public infrastructure
contract.

Output:
Entity: ABC Holdings
Risk Category: Bribery / Corruption
Severity: High
Sentiment: Negative
Confidence: 91%
Explanation: The article mentions an investigation involving bribery in a public contract.
```

11.6 Functional Requirements

ID	Requirement	Priority
AI-001	System should classify adverse media articles by risk category.	Must Have
AI-002	System should extract named entities from article text.	Should Have
AI-003	System should generate short article summaries.	Must Have
AI-004	System should assign severity to adverse media records.	Must Have
AI-005	System should explain why an article is considered risky.	Must Have
AI-006	System should identify low-risk or irrelevant media.	Should Have

11.7 AI Guardrails

- AI should not make final compliance decisions independently.
- AI should not invent evidence.
- AI should not classify a case as a true hit without supporting data.
- AI should show confidence or uncertainty where applicable.
- AI should refer uncertain cases to human review.
- AI should provide source references for important claims.

12. RAG Requirements

12.1 Purpose

The RAG assistant should help analysts ask natural-language questions about a client or case and receive grounded answers using retrieved evidence. The assistant should retrieve relevant information from stored documents and structured case data before generating an answer.

12.2 Knowledge Base

- Client profiles
- Screening hits
- Ownership details
- Board member profiles
- Country risk notes
- Adverse media articles
- Previous case history

- Analyst notes
- Policy-like reference documents
- Generated reports

12.3 RAG Workflow

```
Analyst Question
    -> Query Understanding
    -> Retrieve Relevant Documents
    -> Rank Evidence
    -> Generate Answer
    -> Show Source Evidence
    -> Suggest Next Action
```

12.4 Example Analyst Questions

- Why is ABC Holdings classified as high risk?
- Is this sanctions hit likely a true match or false positive?
- What are the main risk drivers for this client?
- Which countries create risk exposure for this client?
- Summarize all adverse media related to this client.
- What evidence supports enhanced due diligence?
- Has this client had previous alerts?

12.5 RAG Answer Requirements

- Direct answer
- Supporting evidence
- Source references
- Confidence level
- Uncertainty or missing information
- Suggested next step

12.6 Example RAG Answer

```
Question:
Why is ABC Holdings classified as high risk?

Answer:
ABC Holdings is classified as High Risk because the system identified three major risk
drivers:
1. A high-severity adverse media article related to bribery.
2. Exposure to one high-risk jurisdiction.
3. One internal alert from a previous review cycle.

The strongest risk driver is the adverse media article mentioning bribery allegations in
a public infrastructure contract.

Confidence: High
Suggested next step: Perform Enhanced Due Diligence and request senior reviewer approval
before continuing the relationship.
```

12.7 Functional Requirements

ID	Requirement	Priority
RAG-001	System should retrieve relevant client documents based on analyst question.	Must Have
RAG-002	System should generate grounded answers using retrieved context.	Must Have
RAG-003	System should show evidence/source references.	Must Have
RAG-004	System should avoid answering when evidence is missing.	Must Have
RAG-005	System should support client-specific Q&A.	Must Have
RAG-006	System should summarize adverse media and case history.	Should Have

13. Analytics Dashboard Requirements

13.1 Purpose

The dashboard should help analysts and managers understand risk exposure, case status, alert volume, investigation performance, and operational bottlenecks.

13.2 Dashboard Page 1: Executive Overview

Area	Elements
KPIs	Total clients screened, total alerts, high-risk clients, critical-risk clients, open cases, closed cases, average risk score, false positive rate, true hit rate, cases pending review.
Charts	Risk level distribution, alert type distribution, case status distribution, monthly alert trend, high-risk client count by country.

13.3 Dashboard Page 2: Case Queue

- Case ID
- Client name
- Hit type
- Risk score
- Risk level
- Case status
- Assigned analyst
- Aging in days
- Recommended action

Filter / Sort Type	Options
Filters	Risk level, hit type, country, analyst, case status, client status, date range.
Sorting	Highest risk score first, oldest case first, critical cases first, newest alerts first.

13.4 Dashboard Page 3: Client 360 View

- Company profile
- Client status
- Business activity
- Country exposure
- Board members
- UBOs
- Screening hits

- Adverse media
- Previous investigation history
- Risk score breakdown
- Recommended action
- RAG assistant
- Report generation button

13.5 Dashboard Page 4: Country Risk Exposure

- Clients by country
- High-risk country exposure
- Critical-risk jurisdictions
- Alert volume by country
- Risk score distribution by geography

13.6 Dashboard Page 5: Analyst Productivity

- Cases assigned per analyst
- Cases closed per analyst
- Average review time
- Pending cases
- Overdue cases
- Escalation rate
- False positive review volume

13.7 Functional Requirements

ID	Requirement	Priority
DA-001	Dashboard should show compliance KPIs.	Must Have
DA-002	Dashboard should show risk distribution.	Must Have
DA-003	Dashboard should show case queue.	Must Have
DA-004	Dashboard should allow filtering by risk level and hit type.	Must Have
DA-005	Dashboard should provide Client 360 view.	Must Have
DA-006	Dashboard should show analyst productivity metrics.	Should Have
DA-007	Dashboard should show country exposure trends.	Should Have

14. Due Diligence Report Requirements

14.1 Purpose

The system should generate a standardized due diligence report that analysts can review, edit, and submit. The report should improve consistency and reduce repetitive manual writing.

14.2 Report Sections

- Client Summary
- Business Activity
- Client Status
- Ownership and Board Review

- Screening Hit Details
- Entity Matching Result
- Adverse Media Review
- Country Risk Exposure
- Previous Case History
- Risk Score Explanation
- System Recommendation
- Analyst Notes
- Reviewer Comments
- Final Decision
- Evidence References

14.3 Example Report Summary

```
ABC Global Holdings is an active corporate client operating in the logistics sector with
business exposure to UAE, Singapore, and Russia. The system identified one high-severity
adverse media article related to bribery allegations and one internal alert from a
previous review cycle. No direct sanctions exposure was identified.

The client received a risk score of 92 and is classified as High Risk. The primary risk
drivers are adverse media exposure, high-risk country association, and prior internal
alert history.

Recommendation: Enhanced Due Diligence and senior compliance review before continuing the
relationship.
```

14.4 Functional Requirements

ID	Requirement	Priority
REP-001	System should generate a due diligence report for each case.	Must Have
REP-002	Report should include risk score breakdown.	Must Have
REP-003	Report should include evidence summary.	Must Have
REP-004	Report should include final recommendation.	Must Have
REP-005	Analyst should be able to edit the report text.	Should Have
REP-006	Report should be exportable as PDF or Word.	Should Have

15. Case Management Requirements

15.1 Case Statuses

Status	Description
New	Alert generated but not reviewed.
Data Enriched	Data has been collected and merged.
Under Review	Analyst is reviewing case.
Need More Information	Additional evidence required.
Escalated	Sent to senior reviewer.
Cleared	No material risk found.
Monitoring Required	Client requires ongoing monitoring.
Restricted	Business activity restricted.
Offboarded	Relationship ended.
Closed	Case completed.

15.2 Case Actions

- Open a new case
- View case details
- Assign case to analyst
- Change case status
- Add analyst notes
- Add reviewer comments
- Mark true hit or false positive
- Escalate case
- Generate report
- Close case

15.3 Functional Requirements

ID	Requirement	Priority
CM-001	System should create case records from screening hits.	Must Have
CM-002	System should track case status.	Must Have
CM-003	System should allow analyst notes.	Must Have
CM-004	System should allow final decision tagging.	Must Have
CM-005	System should track case aging.	Must Have
CM-006	System should support escalation status.	Should Have

16. Human-in-the-Loop Requirements

16.1 Purpose

Compliance decisions require human judgment. The platform should assist analysts but not replace them. System recommendations and final decisions should be clearly separated.

16.2 Human Review Points

- Risk level is High or Critical.

- Entity match is Possible Match.
- DOB or country data is missing.
- Adverse media severity is High or Critical.
- RAG assistant reports insufficient evidence.
- Analyst overrides system recommendation.
- Final action is Restrict or Offboard.

16.3 Analyst Override

Analysts should be able to override risk classification, match classification, recommended action, or case status. Every override should require a reason, analyst name, timestamp, and supporting note.

16.4 Functional Requirements

ID	Requirement	Priority
HITL-001	System should allow analyst review before final decision.	Must Have
HITL-002	System should require reason for overrides.	Should Have
HITL-003	System should flag cases requiring senior review.	Must Have
HITL-004	System should separate system recommendation from final decision.	Must Have

17. Model and Analytics Evaluation

17.1 Purpose

The platform should measure whether the analytics system improves decision-making, prioritization, report consistency, and operational efficiency.

17.2 Evaluation Metrics

Category	Metrics
Risk Classification Metrics	Accuracy, precision, recall, false positive rate, false negative rate.
Entity Matching Metrics	Match accuracy, false match rate, missed match rate, average similarity score by outcome.
Operational Metrics	Average case review time, cases reviewed per analyst, cases escalated, cases closed, overdue cases.
Business Impact Metrics	Reduction in false-positive review effort, faster identification of high-risk clients, improved report consistency, better risk visibility.

17.3 Baseline Comparison

Manual Process	Automated-Assisted Process
Analyst checks multiple Excel files.	System creates unified client profile.
Manual name comparison.	Fuzzy matching score and explanation.
Subjective prioritization.	Risk-based case queue.
Manual report writing.	Auto-generated report draft.
Limited management view.	Risk analytics dashboard.

17.4 Success Criteria

- Consolidates at least 5 data sources.
- Generates risk scores for all clients.
- Classifies clients into risk levels.
- Identifies top high-risk clients.

- Provides explainable risk score breakdowns.
- Answers client-specific questions using RAG.
- Generates a due diligence report.
- Shows dashboard KPIs and trends.

18. Non-Functional Requirements

Area	Requirement
Usability	The system should be simple enough for a non-technical analyst to upload data, view risk profiles, filter case queues, ask questions, and generate reports.
Explainability	The system must explain why a client is high risk, which indicators contributed to the score, which documents were used by RAG, why a hit is likely true or false positive, and what information is missing.
Data Privacy	Use synthetic or public data only, avoid real client names, avoid confidential bank information, and clearly state the project is for educational/demo purposes.
Reliability	Handle missing files gracefully, show validation errors, avoid crashing on incomplete records, and support reprocessing after data correction.
Performance	Process 1,000 clients within a few minutes, load dashboard views quickly, and return RAG responses within reasonable time for an MVP.
Auditability	Store risk score breakdown, case status changes, analyst overrides, report timestamp, evidence used for RAG answer, and final decision.

19. Technical Approach

19.1 Recommended Stack for Non-Coder / MSBA Project

Layer	Recommended Tool
Data preparation	Python Pandas in Google Colab
Data storage	CSV / Excel / SQLite
Dashboard	Power BI, Tableau, or Streamlit
Entity matching	RapidFuzz
NLP	spaCy, scikit-learn, or LLM-assisted classification
RAG	LangChain or LlamaIndex with FAISS or Chroma
Report generation	Python docx / PDF template
Presentation	PowerPoint / PDF case study

19.2 Simplified Architecture

```
CSV / Excel Data
    -> Python Data Processing Notebook
    -> Cleaned Master Dataset
    -> Risk Scoring Engine
    -> Dashboard
    -> RAG Assistant
    -> Report Generator
```

19.3 Optional Advanced Architecture

```
Frontend Dashboard
    -> Backend API
    -> Data Processing Layer
    -> Database
    -> Risk Scoring Engine
    -> Vector Database
    -> RAG Assistant
    -> Report Generator
```

Recommended direction for MSBA

For the MSBA project, use the simplified architecture. The strongest deliverable is a clear analytics workflow with high-quality data preparation, explainable scoring, dashboards, and a RAG demo.

20. Key Screens and Pages

Screen / Page	Purpose	Main Elements
Home Page	Introduce project and upload datasets.	Project title, upload data button, data quality status, number of clients loaded, number of alerts loaded, process data button.
Dashboard Page	Show overall compliance risk.	KPI cards, risk distribution chart, alert type chart, case status chart, country risk chart, high-risk clients table.
Case Queue Page	Prioritize analyst review.	Case table, filters, risk score, hit type, case aging, recommended action, assigned analyst, case status.
Client 360 Page	Show complete view of one client.	Client profile, ownership, screening hits, adverse media, country exposure, risk score breakdown, RAG Q&A, report generation.
RAG Assistant Page	Allow analyst to ask evidence-backed questions.	Search box, suggested questions, AI response, evidence section, source references, confidence level.
Report Page	Generate and review due diligence report.	Report preview, editable analyst notes, final recommendation, export button.

21. User Stories and Acceptance Criteria

ID	User Story	Description	Acceptance Criteria
US-001	Upload and Process Data	As a compliance analyst, I want to upload multiple due diligence datasets so that the system can create a unified client profile.	User can upload client, ownership, hits, country, adverse media, and case history files. System validates columns, shows missing field errors, creates a merged dataset using client_id, and displays a data quality summary.
US-002	View Client Risk Profile	As an analyst, I want to view all risk-related information for one client so that I do not need to check multiple files manually.	Client profile shows company details, board and ownership information, screening hits, adverse media, country exposure, risk score, and recommendation.
US-003	Identify Possible True Hits	As an analyst, I want the system to compare screening names with internal client names so that I can identify possible true matches.	System calculates similarity score, classifies hit, shows explanation, and allows analyst override.
US-004	Prioritize High-Risk Cases	As a compliance manager, I want cases ranked by risk so analysts can review the most critical cases first.	Cases are ranked by risk score, high-risk cases appear first, filters are available, and overdue cases are visible.
US-005	Ask RAG-Based Questions	As an analyst, I want to ask questions about a client so I can quickly understand risk drivers and evidence.	User can ask questions, system retrieves relevant documents, provides grounded answer, shows source evidence, and highlights missing information.
US-006	Generate Due Diligence Report	As an analyst, I want to generate a report draft so I can save time and maintain report consistency.	System generates a report with client summary, risk indicators, hit analysis, score explanation, recommendation, and export/copy option.

22. MVP Deliverables

- Synthetic dataset.
- Data cleaning notebook.
- Unified client risk dataset.
- Entity matching logic.
- Risk scoring model.
- Dashboard with KPIs and charts.
- Client 360 view.
- RAG assistant demo.

- Automated due diligence report.
- Final presentation explaining business impact.

23. Project Roadmap

Phase	Activities	Output
Phase 1: Business and Data Design	Define workflow, create synthetic datasets, define risk factors, define scoring model, define dashboard KPIs.	Data dictionary, sample data, process flow diagram.
Phase 2: Data Mining Pipeline	Load files, clean data, merge data, create Client 360 dataset, generate data quality report.	Cleaned master dataset, data quality summary.
Phase 3: Risk Analytics	Build entity matching, risk scoring, risk levels, recommendations, and evaluation against synthetic outcomes.	Risk scored dataset, case priority list.
Phase 4: Dashboard	Build KPI dashboard, case queue, Client 360 page, and country exposure analysis.	Interactive dashboard.
Phase 5: AI and RAG	Create knowledge base, embed documents, build retrieval pipeline, generate grounded answers, show evidence.	RAG investigation assistant.
Phase 6: Report and Storytelling	Generate report template, create sample case reports, prepare MSBA-style presentation, document business impact and limitations.	Due diligence report, portfolio case study, presentation deck.

24. Success Metrics

24.1 Product Success Metrics

Metric	Target
Data sources consolidated	5+
Clients processed	500+
Risk score coverage	100% of clients
Cases prioritized	100% of alerts
Report generation	Available for every case
RAG answer grounding	Evidence shown for every answer
Dashboard coverage	Executive + analyst views

24.2 Business Impact Metrics

Metric	Expected Improvement
Manual data lookup effort	Reduced
Case prioritization quality	Improved
Report consistency	Improved
High-risk case visibility	Improved
False positive review efficiency	Improved
Managerial risk oversight	Improved

24.3 Academic / Portfolio Metrics

- Data mining from multiple sources
- Structured analytics pipeline
- Explainable risk scoring
- AI/NLP usage
- RAG implementation
- Dashboard storytelling

- Business decision support

25. Risks and Mitigations

Risk	Impact	Mitigation
Synthetic data may look unrealistic.	Project may feel weak.	Create realistic scenarios, case histories, ownership links, and alert patterns.
AI may hallucinate.	Compliance trust issue.	Use RAG with source evidence and avoid unsupported conclusions.
Risk scoring may look arbitrary.	Weak analytics credibility.	Clearly explain weights, assumptions, and sensitivity.
Project may become too technical.	Hard for MSBA positioning.	Focus on business value, dashboarding, and decision support.
RAG may be difficult to implement.	Delays MVP.	Build a simple document-search demo first.
Too many features.	Scope creep.	Start with MVP only.
No real compliance validation.	Limited realism.	Mention clearly as a project limitation.

26. Assumptions

- The project is for educational and portfolio purposes.
- The user will use synthetic or public data.
- The platform is not used for actual regulatory decision-making.
- The user is not building a production enterprise system.
- The primary goal is MSBA readiness and analytics storytelling.
- The implementation should be manageable for a non-coder.
- Final compliance decisions remain with human reviewers.

27. Open Questions

- Will the dashboard be built in Power BI, Tableau, or Streamlit?
- Will the data pipeline be built in Python notebook or a simple app?
- Will RAG use OpenAI API, local embeddings, or a no-code AI tool?
- What size should the synthetic dataset be?
- Should the report be generated as PDF, Word, or dashboard text?
- Should adverse media data be manually created or generated synthetically?
- Should the project include a scenario-analysis module?
- Should the final presentation focus on compliance, analytics, or AI?

28. Recommended MVP for MSBA

```
Synthetic data
    -> Data mining and cleaning
    -> Unified client risk profile
    -> Entity matching
    -> Explainable risk scoring
    -> Dashboard insights
    -> RAG-based investigation support
    -> Due diligence report
```

The strongest version should emphasize business problem clarity, data mining, explainable risk scoring, dashboard storytelling, case prioritization, and RAG as an advanced feature.

29. Final Product Positioning

Positioning Statement

The AI-Powered Due Diligence & Compliance Risk Intelligence Platform uses data mining, risk analytics, entity matching, dashboarding, and RAG-based AI assistance to help compliance analysts consolidate client information, identify risk indicators, prioritize cases, explain risk decisions, and generate evidence-backed due diligence reports.

This makes the project relevant for MSBA applications, business analytics portfolios, risk analytics roles, compliance analytics roles, RegTech projects, AI for business decision-making, and data-driven operational improvement.

30. Appendix: Glossary

Term	Meaning
CDD	Customer Due Diligence. Review of customer information to understand identity, activity, and risk.
EDD	Enhanced Due Diligence. Deeper review for higher-risk clients or alerts.
AML	Anti-Money Laundering. Controls used to prevent and detect money laundering risks.
KYC	Know Your Customer. Process of verifying and understanding customer identity and risk.
PEP	Politically Exposed Person. Person with political exposure or close association.
UBO	Ultimate Beneficial Owner. Person who ultimately owns or controls an entity.
RAG	Retrieval-Augmented Generation. AI method that retrieves relevant documents before generating an answer.
False Positive	Alert that appears relevant initially but is not actually related after investigation.
True Hit	Alert that is confirmed to be relevant to the client or associated entity.